DHRUV SUTHAR

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EDUCATION

BSc Computer Science (CGPA: 8.55/10)

Shree L.R. Tiwari Degree College of Arts, Commerce & Science, Mumbai (Mira Road), Completed May 2026

Master of Computer Application, Specialization in AI & ML (Part-time)

D. Y. Patil University, Mumbai, 2026 to 2028 (Ongoing)

EXPERIENCE

Software Development Intern, Founding Intern

NStechX India Pvt. Ltd | Nov 2025 – Feb 2026

- Built and deployed reconciliation systems for ATM and IMPS transaction channels, processing high-volume financial transactions per day with 90 to 92% accuracy improvement.
- Designed and shipped an AI-driven Fraud Risk Management System (FRMS) prototype end-to-end (backend anomaly-detection pipeline and frontend dashboard), reducing manual review time/false positives by 85 to 95%.
- Enhanced UPI reconciliation workflows, improving matching accuracy and supporting scalability to 5K transactions.
- Contributed to AWS deployment of backend transaction-processing services and presented product prototypes directly to banking stakeholders.

PROJECTS

LEXICA: Text-to-SQL with RAG and Security Guardrails

Python, FastAPI, React, SQLAlchemy, ChromaDB, Sentence Transformers, Groq, PostgreSQL, SQLite

- Built an AI-powered Text-to-SQL application using Groq LLM and schema-aware RAG to convert natural language questions into SQL queries, retrieving relevant tables, columns, and examples with foreign-key expansion for multi-table joins.
- Developed AST-based SQL validation using SQLGlot to enforce read-only SELECT queries, restrict table and column access, block PII and dangerous functions, and apply automatic row limits.
- Implemented secure query execution through a separate read-only database connection with statement timeouts and result caps; achieved 10/10 successful schema retrievals across test questions.

ANALYTICA: CSV AI Studio

React, Vite, Express.js, Groq (Llama 3), Recharts, PapaParse, Framer Motion

- Built a full-stack CSV analytics platform with auto-profiling (column types, missing values, data quality checks) that updates insights live on upload.
- Developed an interactive Chart Studio with AI-suggested visualizations and natural-language Q&A over uploaded data, powered by a local inference engine with Groq/Llama 3 fallback.
- Hardened the platform with CSRF protection, rate limiting, and file validation to make it production-safe for public use.

JURIX: AI Courtroom Simulation System

NLP, OCR, Multi-Agent Systems

- Presented at Avishkar, a university-level project competition with participation from multiple colleges, and was well-received by the evaluation panel.
- Developed a multi-agent courtroom simulation (Prosecutor, Defense, Judge agents) for legal education, enabling realistic AI-driven case argumentation.
- Built an OCR + NLP pipeline to parse legal documents and feed structured case data into the agent reasoning system.

SKILLS SUMMARY

AI/ML: RAG, Multi-Agent Systems, Prompt Engineering, Gemini AI, Vertex AI, OpenAI, OpenRouter, Ollama

Backend & Databases: FastAPI, Node.js, Python, MongoDB, REST APIs, Postman

Cloud & DevOps: AWS, Git, GitHub Actions, Vercel, Render

ML Frameworks: PyTorch, Scikit-learn, Hugging Face Transformers, spaCy

Core Concepts: Workflow Automation, NLP, Speech AI (STT/TTS), OpenCV, OCR

LEADERSHIP & ACTIVITIES

- CS Department Representative, Carpediem 2026, contributed to PR, coordination, and execution; helped secure the Best PR Trophy.
- Core Committee Member, TechNova 2k25, led coordination of multiple technical events, managing logistics and on-ground execution.
- Technical Team Member, Youth Festival 2024, handled sound systems, stage setup, and technical coordination.